

# Yashwardhan Singh

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## EDUCATION

### Bhagwan Parshuram Institute of Technology, GGSIPU

Delhi, India

*B.Tech in Information Technology*

*Aug. 2024 – Aug. 2028*

- Active member of Namespace BPIT - tech society focused on software development, open source, and competitive programming.

## PROJECTS

### AlgoHub — LeetCode-Style Code Evaluation Platform

*Node.js · TypeScript · Fastify · BullMQ · Redis · Docker · MongoDB · React* | [GitHub](#)

- Architected a distributed online judge across 4 decoupled microservices: Enqueuer, Evaluation, Problem, and WebSocket services — each independently deployable.
- Built a language-agnostic container factory (C++, Python, Java) using Dockerode to safely sandbox and execute user code against hidden test cases.
- Implemented async submission pipeline using BullMQ (Redis-backed) with separate submission and evaluation queues; integrated Bull Board for real-time queue monitoring.
- Used Fastify for the Enqueuer API with Zod validation, DTOs, and repository pattern; dedicated WebSocket service pushes real-time evaluation results to clients.
- Built a React + TypeScript frontend with resizable split-panel code editor (Ace Editor), multi-language support, real-time verdict display via WebSocket context, and Markdown-rendered problem statements.

### Adaptive RAG Application (WIP)

*Python · FastAPI · LangGraph · LangChain · Qdrant · Redis · MongoDB · OpenAI · React* | [GitHub](#)

- Architected a multi-step agentic RAG pipeline using LangGraph with an explicit execution graph: pre\_planner → query rewrite → retrieve → evaluator → rerank → generate; each node independently testable.
- Built a pre-retrieval planner that classifies query intent and dynamically routes to single-query or multi-query rewrite strategies, improving vector retrieval precision.
- Implemented confidence-based routing in a post-retrieval evaluator node that scores context relevance and falls back to direct LLM generation when retrieval quality is insufficient.
- Added cross-encoder reranking (dedicated reranker model) to re-score Qdrant top-k results with cosine similarity scores surfaced in the UI via a DocPanel component.
- Integrated Redis-backed semantic and full-response caching with async cleanup, eliminating redundant LLM calls; used MongoDB as a LangGraph checkpoint for conversation state persistence.

### Scalable Authentication Service

*Node.js · Express.js · MongoDB · Argon2 · JWT · Nodemailer* | [GitHub](#)

- Built a full authentication microservice following a clean Routes → Controller → Service → Repository layered architecture for separation of concerns and maintainability.
- Implemented multi-device session management with per-device refresh token rotation, HTTP-only cookie delivery, and full session revocation (single + all devices).
- Secured with Argon2 hashing (passwords, OTPs, refresh tokens), Email/OTP verification via Nodemailer + OAuth2, MongoDB TTL auto-expiry, and tiered rate limiting across 9 endpoints.

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, Python, C++

**Backend & Frameworks:** Node.js, Express.js, Fastify, FastAPI, REST API Design, JWT Auth, Microservices

**AI & ML:** LangChain, LangGraph, RAG Pipelines, Prompt Engineering, LLM Integration, OpenAI API

**Databases & Caching:** MongoDB, Redis, Qdrant (Vector Database)

**Frontend:** React, Tailwind CSS

**DevOps & Tools:** Docker, Git, GitHub, Linux